

DOMINIC CARRILLO

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Biography

In final year of Ph.D. candidacy in Computer Science and Engineering at the University of North Texas, conducting research in Autonomous Systems with emphasis on Computer Vision, Object Detection, Multi-Sensor Fusion, Registration, and Simultaneous Localization and Mapping (SLAM) under the supervision of Dr. Qing Yang. My work focuses on advancing cooperative and connected automated vehicle perception to support safer, more efficient, and bandwidth-aware transportation systems.

Passion for teaching and mentoring, especially in mentoring students in both basic and advanced areas of computer science. Conduct courses such as Linux systems and introductory and senior-level CS courses designed to prepare students for real-world engineering challenges. My goal beyond the classroom is to mentor students by linking them to meaningful projects, research opportunities, and industry-relevant experiences that promote their development and transition into professional roles with confidence.

In addition, have an ambition to continue to develop and continue my research in autonomous vehicle perception and cooperative robotics. Having an active research agenda will help me in my long-term transition to an assistant professorship where I can help mentor students who want to become researchers and contribute to impactful work in robotics, perception, and intelligent transportation systems.

Education

PhD. Computer Science and Engineering†	Aug. 2019 - Expect Dec. 2026
University of North Texas, Denton, TX, USA	
B.S. Computer Science and B.S. Mathematics‡	Aug. 2015 - Dec. 2018
Sul Ross State University, Alpine, TX, USA	
International Summer School Program	July 2018 - Aug. 2018
Yonsei International Summer School, Seoul, Korea	
General Studies	Aug. 2014 - May 2015
Midland College, Midland, TX, USA	

Publications

Journals

- [J1] **Dominic Carrillo**, Michael Nutt, Maarten Meijer, Junaid Khan, Song Fu, Qing Yang. "*Bringing Different Views Together: A Hybrid Cooperative Perception Framework for Connected Autonomous Vehicles*". IEEE Network. 2025.
- [J2] Jingda Guo, **Dominic Carrillo**, Qi Chen, Qing Yang, Song Fu, Hongsheng Lu, Rui Guo. "*Slim-FCP: Lightweight Feature-Based Cooperative Perception for Connected Automated Vehicles*". IEEE Internet of Things Journal. 2022.
- [J3] Jingda Guo, **Dominic Carrillo**, Sihai Tang, Qi Chen, Qing Yang, Song Fu, Xi Wang, Nannan Wang, Paparao Palacharla. "*CoFF: Cooperative Spatial Feature Fusion for 3-D Object Detection on Autonomous Vehicles*". IEEE Internet of Things Journal. 2021.

Conferences

- [C1] Deyuan Qu, **Dominic Carillo**, Qing Yang, and Song Fu, "*Location Privacy Protection and 911 Task Allocation in Vehicle-Based via Differential Privacy*". IEEE Vehicular Technology Conference (VTC2024-Fall). 2024.
- [C2] Dhakal, Sudip, Qi Chen, Deyuan Qu, **Dominic Carillo**, Qing Yang, and Song Fu, "*Sniffer Faster R-CNN: A Joint Camera-LiDAR Object Detection Framework with Proposal Refinement*". IEEE International Conference on Mobility, Operations, Services and Technologies. 2023.
- [C3] Sudip Dhakal, Deyuan Qu, **Dominic Carrillo**, Qing Yang, Song Fu. "*OASD: An Open Approach to Self-Driving Vehicle*". MetroCAD. 2021.

Research

Lab & Partnerships

- [R1] **IUCRC Center for Electric, Connected and Autonomous Technologies for Mobility (eCAT)†**
Jan. 2024 - CURRENT
Engaged in collaborative research efforts, focusing on the development and deployment of next-generation technologies for electrification, resilient sensing/perception, and real-time edge computing. My role involves active communication and knowledge-sharing with partner universities, where we exchange insights to advance safety and efficiency in connected autonomous vehicles. While my contributions are aligned with my broader research interests, this collaboration has enriched my understanding of diverse approaches and innovations within the field.
- [R2] **Connected Autonomous Vehicles Lab†**
Aug. 2019 - CURRENT
Conducted extensive application-based research in autonomous vehicle technologies covering 2D/3D data fusion, deep learning models, cooperative perception, and edge computing among other important domains. Using Autoware and ROS to validate our ideas, I have been crucial in designing and testing these systems on our real-world platforms such as the Polaris GEM vehicle. This practical experience has improved my technical proficiency and put me in a position to make a significant contribution to the development of autonomous vehicle technology, with an emphasis on how these advancements can be used to benefit marginalized communities and increase public safety.
- [R3] **SuperTruck3 with PACCAR (Under NDA)†**
Jan. 2023 - Dec. 2025
This is a U.S. Department of Energy initiative aimed at developing advanced technologies to significantly improve the efficiency and sustainability of heavy-duty trucks. As part of this project, my role includes informing our industry partners about the capabilities of connected autonomous vehicles, particularly how these technologies can be integrated to achieve the project's goals of optimizing fuel efficiency, reducing carbon emissions, and reshaping the future of the trucking industry.
- [R4] **F1TENTH Environment System in CIIMS Lab†**
June 2020 - May 2022
Oversaw the building of the F1TENTH system under the CIIMS project, an open-source platform intended for research and education on autonomous vehicles. Research in reinforcement learning, robotics, and communication systems has made use of such flexible framework. Beyond my regular developing work, I continue to assist others and provide TAMS and undergraduate computer science and computer vision students with a valuable learning tool. Through means of this platform, I guide

students in acquiring practical knowledge of innovative technologies and therefore stimulate their enthusiasm in the fast-developing field of autonomous systems.

Projects

[P1] **DeepAlign: Training-Free Occupancy Registration as a Replacement for Oracle Pose in Cooperative 3D Detection**[†]

Jan. 2026 - CURRENT

Developed a two-stage pipeline for multi-agent 3D detection in which each vehicle condenses local 3D detection into a fixed-size BEV occupancy grid (OccGrid), followed by a training-free registration module (OccPAM) that computes SE(2) transforms using object-graph matching and robust geometric estimation. Integrated DeepAlign into OpenCOOD as a replacement for oracle CAV-to-ego pose in intermediate fusion, enabling practical alignment without GNSS, IMU, or shared pose metadata. Evaluated seven intermediate fusion architectures on 2,170 OPV2V test frames, quantifying the detection performance gap between oracle pose and DeepAlign-estimated alignment.

[P2] **Camera & Radar Fusion for Perception with Bandwidth Efficiency**[†]

Feb. 2025 - Dec. 2026

Developed a camera-radar fusion pipeline that use radar-based segmentation to segment object-centric regions and provide depth cues to the camera data stream, based on the CRN architecture, therefore enhancing the reliability of BEV feature generation for autonomous perception. Evaluated the impact of fused depth-enriched features on detecting objects with AP performance and analyzed compressed or selective transmission methods to facilitate bandwidth-efficient cooperative perception.

[P3] **Bringing Different Views Together: A Hybrid Method for Cooperative Perception in Autonomous Vehicles**[†]

Jan. 2024 - Feb. 2025

A hybrid cooperative perception fusion system for connected autonomous vehicles is shown in this article with the objective of reducing the high transmission costs associated with real-time data. The framework improves data transmission based on available bandwidth or data augmentation methods, ensuring vehicles can rapidly communicate high-confidence data without saturating the network. Experimental findings indicate that this strategy considerably diminishes data transmission sizes while preserving object detection accuracy, offering a scalable and efficient option for cooperative perception in autonomous vehicle systems.

[P4] **FR-Cooper: Feature Map Region based Cooperative Perception using 3D Point Clouds**[†]

Aug. 2023 - May 2024

In the context of autonomous vehicles, effective data sharing between vehicles is essential, but network bandwidth restrictions provide a barrier. The current method is to transmit these feature maps, which can be large, between vehicles. We developed the FR-Cooper approach to overcome this restriction while maintaining object association. FR-Cooper extracts specified object regions out of the feature map to minimize data transmission. Only these relevant object regions will be sent between vehicles rather than the complete map. This work merged into the next work [P3] as a section by discussing multiple cooperative fusion methods.

[P5] **Raw Data Cooperative Perception on Autonomous Vehicles**[†]

Feb. 2021 - May 2021

First preliminary study utilizing the F1TENTH Environment. Presented this work at 2021 CPS-IoT Week for Student Design Competition Networked Computing on the Edge which I have received award of \$200 for an Honorable Mention on the Presentation. This study utilizes ROS to

demonstrate that cooperative perception allows a vehicle to benefit from other vehicles to detect objects that are obscured from itself.

[P6] Comprehensive Analysis of TEASER and NDT for Autonomous Vehicle Applications†

Aug. 2021 - Dec. 2021

Evaluated algorithms performance between TEASER and NDT registration utilizing MATLAB and C++ to compare both algorithms, which have claimed to be efficient and have a low computation over ICP. However, there is no correlated research of the two approaches to demonstrate their performance between each other. The project goal is to conduct an experiment implementation of TEASER and the NDT algorithm on our vehicle.

[P7] The Impact of Accommodated Vehicle to Vehicle Communication‡

June 2017 - Oct. 2017

Study of Vehicle-to-Vehicle communication to verify the response of vehicles to decrease accidents on roadways. Vehicles can relay information to each other which allows other vehicles to prepare to stop or respond with the correct course of action. The simulation is built in Unity to have repeatability to showcase the impact of this communication system.

[P8] Creation of a Perfect Hockey Bracket: Use of Matrices to Depict the Outcome of the Stanley Cup‡

June 2016 - Oct. 2016

Input National Hockey League statistical data into the Colley Method to manipulate matrices to rank the teams. From the ranks collected we deduce who is the potential 2016 Stanley Cup Champion and compare it to actual results.

Poster, Presentations, Talks

- *Honorable Mention for Presentation* in CPS-IoT Week - D. Carrillo, D. Qu, S. Dhakal, S. Tang, and Q. Yang, "Raw Data Cooperative Perception on Autonomous Vehicles," CPS-IoT Week 2021 Student Design Competition on Networked Computing on the Edge, Virtual Conference, May 2021

Professional Activities

Reviewer

- IEEE International Conference on Mobility: Operations, Services, and Technologies (MOST), 2024.
- IEEE International Conference on Computer Communications (INFOCOM), 2023.

Graduate Representative Officer †

Society Hispanic of Professional Engineers

Apr. 2021 - May 2025

- Support undergraduates to improve in their professional careers by giving them advice on things like job planning, resume writing, and how to network.
- Promote students to consider applying to graduate school inviting faculty to present their background and explain to students that obtaining an advanced degree will open doors to more possibilities.
- Facilitate communication with SHPE alumni, establishing ties that can help current students with mentorship and professional development, through social events.
- Motivate students to complete their academic pursuits by offering resources, and assistance.
- Collaborate with other student leaders to support with events and initiatives, including moderating the SHPE UNT Discord server, assisting in building and maintaining the SHPE UNT website, and actively participating in events to enhance community engagement.

Teaching Experience

TF CSCE 1032 † - Introduction to Computer Programming Fall 2025

Supported instruction for an introductory Python programming course by delivering lectures based on the department curriculum and enhancing course materials with updated slides and examples. Utilized Google Colab as the primary programming environment to help students practice Python interactively throughout the semester. Guided students through weekly labs using zyLabs, providing assistance, clarification, and debugging support as they worked through structured exercises and programming problems. Mentored students individually and in groups to strengthen foundational programming skills and build confidence in applying computational thinking to real-world tasks.

TA CSCE 4600 † - Introduction to Operating System Spring 2021

TA CSCE 1010 † - Discovering Computer Science Fall 2019

Assisted in administering courses and recitations with responsibilities in grading assignments, leading discussions, and providing one-on-one support to students. This educational endeavor is one of many that has enabled me to be proficient in explaining complicated ideas clearly and has sparked my desire to teach and guide, even in the field of computer science.

Previous Work Experience

Electrical Mechanic and Shop Hands June 2014 - July 2019

Carrillo's Automotive, Midland, TX

Conducted diagnostic inspections on electrical issues for vehicles, including electrical break tracing, resistance measurement reading, and computer troubleshooting. Developed strong problem-solving and attention-to-detail skills that have helped me in my education, especially when it comes to fixing complex problems.

Awards

- *Awarded* the Google-CAHSI Dissertation Award at the GMIS Conference in Fort Worth on Nov. 8th, 2024.
- *Honorable Mention for Exploration* in CPS-IoT Week - A. Liu, O. Xie, D. Carrillo, and Q. Yang, "Phantom Traffic Jam Prevention Using Cooperative Car Following Model," CPS Week 2022 Student Design on Competition Networked Computing on the Edge, Virtual Conference, May 2022
- *Honorable Mention for Presentation* in CPS-IoT Week - D. Carrillo, D. Qu, S. Dhakal, S. Tang, and Q. Yang, "Raw Data Cooperative Perception on Autonomous Vehicles," CPS-IoT Week 2021 Student Design Competition on Networked Computing on the Edge, Virtual Conference, May 2021

Technical Skills

Program Language - C# | C++ | Python | MATLAB | LaTeX | HTML | CSS

Computer Tools - ROS | Git | Unity | PyCharm | Vim | VirtualBox

Language - Fluent English | Basic Spanish | Elementary Korean